

EDUCATION

■ University of Bradford

Exchange Research Student

Jan. 2025 – Jul. 2025

Thesis: Optimization of Multistage Humidification and Dehumidification Desalination Plant Using Solar Energy (Experimental and Theoretical Study)

■ Politecnico di Milano

M.Sc. Chemical Engineering — Energy and Environment Track

Sept. 2023 – Present

■ Sharif University of Technology

B.Sc. Chemical Engineering

Sept. 2019 – Jul. 2023

Thesis: Simulation of gaseous pollutant conversion on a TWC with a honeycomb structure

EXPERIENCE

Research Institute of Petroleum Industry

Research Intern — Supervisor: Prof. Alimorad Rashidi

Jul. 2022 – Sept. 2022

- Synthesized Cordierite phase and studied thermal operation of monoliths for TWC applications
- Achieved optimal monolith composition through neural network modelling
- Utilised XRD, XRF, SEM, and TEM techniques for material characterisation

Iran's National Elites' Foundation

Research Assistant

Jan. 2022 – Sept. 2022

- Investigated electrochemical treatment methods for removing nitrate, lead, and copper from industrial wastewater
- Developed and tested efficient electrochemical removal processes

Sharif University of Technology

Teaching Assistant

• Kinetics and Reactor Design

Instructor: Prof. Farhad Khorashe

Feb. 2023 – Jun. 2023

• Kinetics and Reactor Design

Instructor: Prof. Mohammad Kazemini

Sept. 2022 – Feb. 2023

• Numerical Methods

Instructor: Prof. Farhad Khorashe

Sept. 2022 – Feb. 2023

• Mass Transfer Operation

Instructor: Prof. Mohammad Kazemini

Feb. 2022 – Jun. 2022

PUBLICATIONS

[1] GhanbariPakdehi, M.S. & Rajaei, A. (2026). *Biomass combustion: introduction and technologies*. Chapter *In Press* in *Encyclopedia of Renewable Energy*, Elsevier.

[2] Ghanbaripakdehi, M. & Kazemini, M. (2023). Investigation on Simulation of Methane Thermal Conversion and its Effective Factors in Syngas. *Proceedings of the 5th International Congress on Chemistry and Chemical Engineering*, Tehran, Iran. [Access Link](#)

ACADEMIC PROJECTS

Prediction of Thermodynamic Properties Based on Functional Groups

May 2025 – Jul. 2025

- Developed and compared classical machine learning algorithms and neural networks to predict thermodynamic properties of molecules based on quantum features. [\[GitHub\]](#)

Techno-Economic Analysis and LCA of Biochar Production

Apr. 2025 – May 2025

- Designed a pilot-scale biochar production plant via slow pyrolysis of corncob waste, optimising the process by utilising syngas for internal heating and sizing key equipment (rotary kiln, dryers)

- Conducted a Class 5 techno-economic analysis to estimate CAPEX and OPEX; calculated NPV, IRR, and ROI; performed sensitivity analysis on feed rates and product pricing
- Performed a cradle-to-gate Life Cycle Assessment (LCA) to quantify GHG emissions (CO₂, CH₄); analysed environmental impacts under various electricity-mix scenarios

Conceptual Design and Economic Feasibility of HDA of Toluene Plant Mar. 2024 – Apr. 2024

- Simulated conceptual design and conducted economic feasibility study of a hydrodealkylation plant using MATLAB and Aspen HYSYS; performed NPV, IRR, ROI, and PBP analysis. [\[GitHub\]](#)

Numerical Methods and Their Application in Chemical Engineering Aug. 2023 – Present

- Comprehensive repository offering educational content, practical code samples, and real-world applications to enhance problem-solving skills. [\[GitHub\]](#)

B.Sc. Thesis — Catalytic Converter Modelling Jul. 2022 – Jul. 2023

Supervisor: Prof. Tayebbeh Hamzehlouyan

- Modelled gaseous pollutant conversion and mass-transfer phenomena in a catalytic converter with a honeycomb structure using COMSOL Multiphysics and MATLAB; analysed the influence of operating parameters on conversion efficiency. [\[Full Document\]](#)

Technical and Financial Analysis of 2-Ethylhexyl Acrylate Plant Jan. 2023 – Feb. 2023

- Designed and simulated a 2-Ethylhexyl Acrylate plant in Aspen HYSYS — reaction, separation, and heat-integration systems
- Performed market study and calculated NPV, IRR, ROI, and PBP over the plant lifetime

PID Controller Design for a CSTR System (Python) Feb. 2023

- Designed a PID temperature controller for a heat-exchanger system using control theory and simulation techniques in Python
- Achieved optimised controller parameters via the Levenberg–Marquardt algorithm. [\[GitHub\]](#)

Flash Distillation Tower Simulation in MATLAB May 2022

- Simulated a flash distillation tower and shell-and-tube heat exchanger using finite-difference methods; calculated heat-transfer coefficients and temperature profiles. [\[GitHub\]](#)

Steady-State Heat Transfer: Finned-Plate Heat Exchanger (MATLAB) Jun. 2021 – Jul. 2021

- Derived temperature equations using the finite-difference method with appropriate boundary conditions; solved linear systems via Gauss–Seidel iteration
- Generated temperature-distribution plots and calculated overall heat loss, validating model accuracy. [\[GitHub\]](#)

TECHNICAL SKILLS

Programming	Python (Pandas, NumPy, SciPy, TensorFlow), MATLAB, gPROMS
Software	COMSOL Multiphysics, Aspen HYSYS, GAMS, Dynsim, Git, Microsoft Office
Soft Skills	Time Management, Teamwork, Problem-Solving, Documentation

VOLUNTEERING EXPERIENCE

Sharif University of Technology — Kimia Scientific Group Mar. 2024 – Apr. 2024

Instructor

- Held a workshop introducing MATLAB to undergraduate Chemical and Petroleum Engineering students. [\[GitHub\]](#)
- Delivered an elementary MATLAB course for the same audience. [\[GitHub\]](#)

AWARDS & SCHOLARSHIPS

Politecnico di Milano International Mobility Grant	2025
Iran's National Elites' Foundation Scholarship to Talented Students	2021, 2022